# Ross Dempsey

Ph.D. Candidate, Princeton University Department of Physics

## Education

- Princeton University — Princeton, NJ
  *Ph.D. Physics* — 2019 – 2025
- Johns Hopkins University — Baltimore, MD
  *BS Physics; BA, MA Mathematics* — 2016 – 2018

## Awards and Honors

- Charlotte Elizabeth Procter Fellowship — 2024 – 2025
  *Princeton University*
- Graduate Research Fellow — 2019 – 2024
  *National Science Foundation*
- First Place — 2019
  *International Theoretical Physics Olympiad*
- Kerr Memorial Prize — 2019
  *Johns Hopkins University*

## Publications

### Papers


- Ross Dempsey et al. "Small Circle Expansion for Adjoint $QCD_2$ with Periodic Boundary Conditions". In: (June 2024). arXiv: 2406.17079 [hep-th]
- Ross Dempsey et al. "Global Symmetry and Integral Constraint on Superconformal Lines in Four Dimensions". In: (May 2024). arXiv: 2405.10914 [hep-th]
- Shai M. Chester, Ross Dempsey, and Silviu S. Pufu. "Level repulsion in $\mathcal{N} = 4$ super-Yang-Mills via integrability, holography, and the bootstrap". In: *JHEP* 07 (2024), p. 059. DOI: 10.1007/JHEP07(2024)059. arXiv: 2312.12576 [hep-th]
- Ross Dempsey et al. "Lattice Hamiltonian for adjoint $QCD_2$". In: *JHEP* 08 (2024), p. 009. DOI: 10.1007/JHEP08(2024)009. arXiv: 2311.09334 [hep-th]
- Ross Dempsey et al. "Phase Diagram of the Two-Flavor Schwinger Model at Zero Temperature". In: *Phys. Rev. Lett.* 132.3 (2024), p. 031603. DOI: 10.1103/PhysRevLett.132.031603. arXiv: 2305.04437 [hep-th]
- Ross Dempsey et al. "Discrete chiral symmetry and mass shift in the lattice Hamiltonian approach to the Schwinger model". In: *Phys. Rev. Res.* 4 (4 Nov. 2022), p. 043133. arXiv: 2206.05308
- Ross Dempsey and Peter Weck. "Compactifying the Kerr-Schild double copy". In: *JHEP* 05 (2023), p. 198. DOI: 10.1007/JHEP05(2023)198. arXiv: 2211.14327 [hep-th]

- Ross Dempsey et al. "Adjoint Majorana $QCD_2$ at finite N". in: *JHEP* 04 (2023), p. 107. DOI: 10.1007/JHEP04(2023)107. arXiv: 2210.10895 [hep-th]
- Shai M. Chester, Ross Dempsey, and Silviu S. Pufu. "Bootstrapping $\mathcal{N} = 4$ super-Yang-Mills on the conformal manifold". In: *JHEP* 01 (2023), p. 038. DOI: 10.1007/JHEP01(2023)038. arXiv: 2111.07989
- Ross Dempsey, Igor R. Klebanov, and Silviu S. Pufu. "Exact symmetries and threshold states in two-dimensional models for QCD". in: *JHEP* 10 (2021), p. 096. DOI: 10.1007/JHEP10(2021)096. arXiv: 2101.05432 [hep-th]
- Kelsey M. Hallinen et al. "Decoding locomotion from population neural activity in moving C. elegans". In: *bioRxiv* (2021). DOI: 10.1101/445643
- Ross Dempsey and Charles Guinn. "A Phase Transition in Minesweeper". In: *10th International Conference on Fun with Algorithms (FUN 2021)*. Vol. 157. Leibniz International Proceedings in Informatics (LIPIcs). 2020, 12:1–12:10. DOI: 10.4230/LIPIcs.FUN.2021.12
- Ross Dempsey, Nadia L. Zakamska, and James E. Owen. "Formation of Orion fingers". In: *MNRAS* 495.1 (May 2020), pp. 1172–1187. DOI: 10.1093/mnras/staa1264. arXiv: 2005.01902
- Ibrahima Bah, Ross Dempsey, and Peter Weck. "Kerr-Schild Double Copy and Complex Worldlines". In: *JHEP* 02 (2020), p. 180. DOI: 10.1007/JHEP02(2020)180. arXiv: 1910.04197 [hep-th]
- J. Chen et al. "The spatial extension of extended narrow line regions in MaNGA AGN". in: *MNRAS* 489.1 (Oct. 2019), pp. 855–867. DOI: 10.1093/mnras/stz2183. arXiv: 1908.02885
- Ross Dempsey and Nadia L. Zakamska. "The size-luminosity relationship of quasar narrow-line regions". In: *MNRAS* 477 (July 2018), pp. 4615–4626. DOI: 10.1093/mnras/sty941. arXiv: 1804.05848

**Books**

- Ross Dempsey and Andrew M. Leifer. *Undergraduate Physics in a Hurry*. Princeton University Press, forthcoming

## Talks

**Invited Research Talks**

- New Approaches to Adjoint $QCD_2$ — July 2024
  *Bootstrap 2024* — UCM (Madrid)
- Adjoint $QCD_2$ on the Hamiltonian Lattice — May 2024
  *Confining Strings* — SwissMAP (Les Diablerets)
- Level Repulsion in $\mathcal{N} = 4$ super-Yang-Mills — May 2024
  *Bootstrap, Localization and Holography* — YITP (Kyoto)
- Phase Diagram of the Two-Flavor Schwinger Model at Zero Temperature — February 2024
  *High-energy Theory Seminar* — Princeton University
- Studying 2D Gauge Theories on the Hamiltonian Lattice — October 2023
  *RBRC Seminar* — Brookhaven National Laboratory

- Fun with the Lattice Schwinger Model — June 2023
  *High-energy Theory Seminar* — Johns Hopkins University

- Phase Diagram of the Two-Flavor Schwinger Model at Zero Temperature — May 2023
  *Quark Confinement 2023* — University of Minnesota

- Adjoint $QCD_2$ at Finite $N$ — September 2022
  *Simons Confinement Collaboration Inaugural Workshop* — Princeton University

**Outreach Talks**

- Physics of the Solar Eclipse
  *Stone Hill Learning Center (Princeton, NJ)* — May 2024

- Quarks, Billiards, and Dualities
  *Warrior-Scholar Project (Princeton, NJ)* — July 2024
  *Stone Hill Learning Center (Princeton, NJ)* — February 2024
  *Warrior-Scholar Project (Princeton, NJ)* — July 2023
  *Sant High School Physics Club (Ulaanbaatur, Mongolia)* — October 2022
  *William & Mary Society of Physics Students (Williamsburg, VA)* — September 2022
  *Da Vinci Science High School (Los Angeles, CA)* — February 2022
  *Warrior-Scholar Project (Princeton, NJ)* — June 2021

- Imaginary Numbers for Real Black Holes — July 2022
  *EDI Summer Research Talks (Princeton, NJ)*

- From Cookie Sheets to Superconductivity — October 2021
  *Hillsborough High School (Tampa, FL)*

**Miscellaneous**

- Mathematica Hogwarts: Becoming a Wizard — January 2024
  *Workshop during Wintersession* — Princeton University

## Teaching

**Princeton**
Assistant in Instruction:

- PHY 208: Principles of Quantum Mechanics — Spring 2022

- PHY 103: General Physics I — Fall 2021

Undergraduate projects co-advised:

- Noah Luch Senior Thesis: "1+1D QCD with Quarks and Adjoint Fermions at Finite $N$"

- George Zhou Senior Thesis: "Entanglement Dynamics in the Schwinger Model"

- George Zhou JP: "Second Order Phase Transition in the Lattice Schwinger Model"

- Loki Lin Senior Thesis: "Discretized Light-Cone Quantization of Two-Dimensional SU($N$) Gauge Theories"

- Alex McDonald JP: "Supersymmetry in DLCQ for Two Dimensional Adjoint QCD"

- Loki Lin JP: "A Two-Dimensional SU($N$) Theory of Quarks"

**Prison Teaching Initiative**

- CSIT-103: Computer Concepts and Literacy — Fall 2022
  *South Woods State Prison*
- CISY-102: Computer Literacy — Spring 2022
  *Edna Mahan Correctional Facility for Women*
- MATH-020: Elementary Algebra — Fall 2021
  *Garden State Youth Correctional Facility*
- MATH-020: Elementary Algebra — Summer 2021
  *Edna Mahan Correctional Facility for Women*
- MAT-033: Pre-algebra — Spring 2020
  *FCI Fort Dix (abbreviated due to COVID)*

## Service

**Princeton**

- Conference for Undergraduate Women in Physics (CUWiP) — August 2022 – January 2023
  *Local Organizing Committee*
- Broader Outreach Committee — 2021 – present
  *"Zoom a Princeton Physicist" talks*
  *Science demonstrations at local schools and libraries*
- Graduate Physics Ambassadors — September 2021 – January 2022
  *Panelist for graduate admissions webinars*
  *Panelist + speaker for Princeton Physics PhD Preview program*

**External**

- Department of Energy National Science Bowl — 2020 – present
  *Competition official, question writer, question editor, Subject Matter Expert for physics*
- Peer Reviewer
  *Physical Review D* — Since 2024
  *Nuclear Physics B* — Since 2024
  *Journal of High Energy Physics* — Since 2023
  *Physical Review Letters* — Since 2022